

SafetySense

George Teifel · Mary Teifel

UNM & CNM APP CONTEST 2026

WHO WE ARE

SafetySense calls for help when you can't.

George Teifel

University of New Mexico

Senior · Electrical Engineering

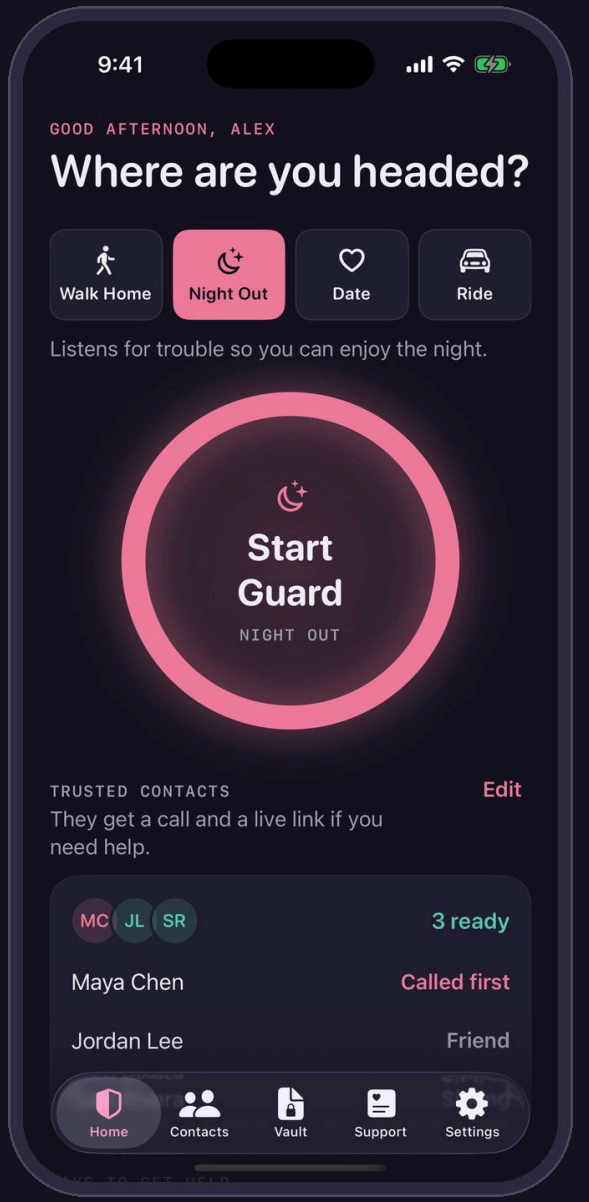
Built the iOS app and the backend

Mary Teifel

Central New Mexico Community College

Freshman · Mechanical Engineering

Led research, UX and the app's wording



THE PROBLEM

Most safety apps wait for you to press a button or set a timer

70%

of 298 women reported significant tonic immobility, an involuntary freeze, during sexual assault

Möller et al., Acta Obstet Gynecol Scand, 2017 (n = 298)

Freezing is involuntary

It is a normal response, not a choice.

Phones get taken, broken or switched off

A dead phone sends nothing.

Speaking isn't always possible

New Mexico is still listed as the only state without text-to-911.

What should a phone do when she can't?

“Most safety apps”: based on the documentation of the apps we reviewed. We did not test them.

OUR ANSWER

You turn on a Guard. If you can't respond, it acts for you.

START IT

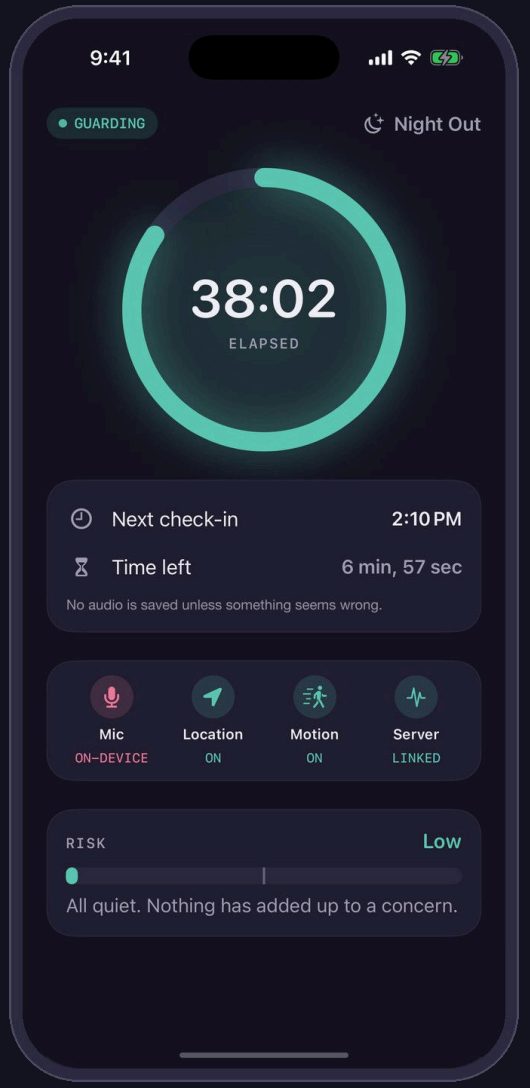
Before a walk home, a night out, a date or a ride, tap Start Guard.

Or skip straight to an alert with the Action Button or Siri.

WHAT IT WATCHES FOR

- **Screams**
- **Breaking glass**
- **Safe word**
- **Fall, then stillness**

It only listens while a Guard is on.



BUILT WITH Native SwiftUI on iOS 26 · Apple on-device sound, speech and motion
Supabase · Twilio · Noonlight (sandbox)

Watch three things.

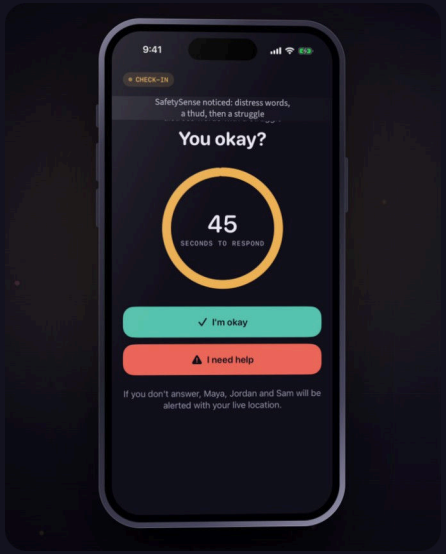
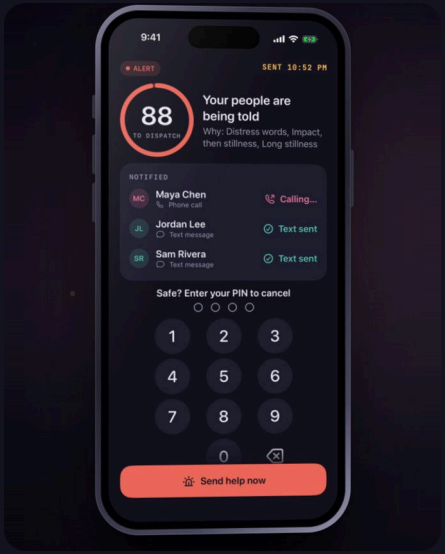
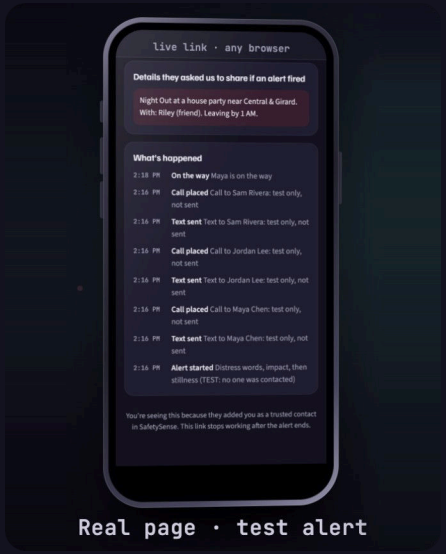
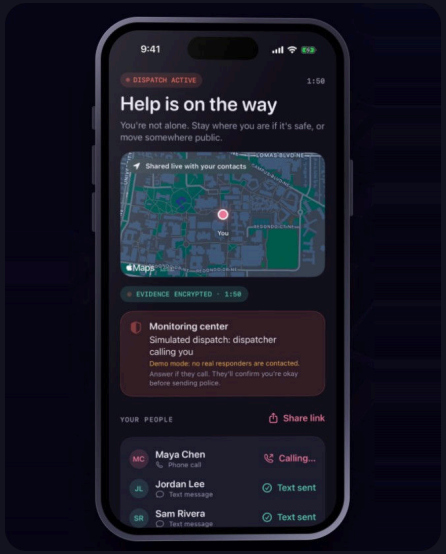
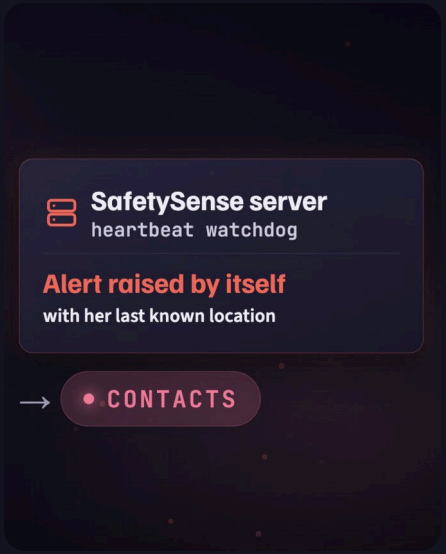
1. It asks before it alerts.
2. What her friend sees, and who's called if no one answers.
3. What happens if someone makes her cancel, or breaks her phone.





WHAT YOU JUST SAW

The app is real. Here's exactly what was simulated.

REAL APP	CALLS WORK	REAL PAGE	SIMULATED	ANIMATION
				
Recorded in the simulator, in demo mode	Real calls in our tests, not in this recording. Texts are sent but don't arrive yet.	Screenshot of a test alert	No police contacted. We test with Noonlight's sandbox	Of a server check we've deployed

Real page · test alert

Waits are sped up, and Ava and Maya are animated characters.
The duress PIN is built; what it reaches is the same simulated dispatch.

ORIGINALITY

Others have pieces. We connected the chain.

	UNM Safe	Noonlight	bSafe	Apple built-in	SafetySense
Notices distress on its own	–	Crash (paid)	Voice phrase	Crash; fall on Watch	✓ **Sound, motion, context**
Acts when the phone goes dark	–	Not documented	Not listed	Check In, timing varies	✓ **Server heartbeat (deployed)**
Duress PIN (fake cancel)	–	–	Not listed	–	✓ **The two PINs you saw**
Professional dispatch	Call UNMPD	✓	–	911 call	✓ **Noonlight (sandbox)**
Live location to a friend	✓	✓	✓	✓ Check In	✓

From each product's documentation; not tested by us. These are four tools a student could already have.
Their columns are shipped products. Ours is a prototype; dispatch runs in Noonlight's sandbox.

DESIGN AND EXPERIENCE

Designed for the worst night of someone's life

Calm words, never blame

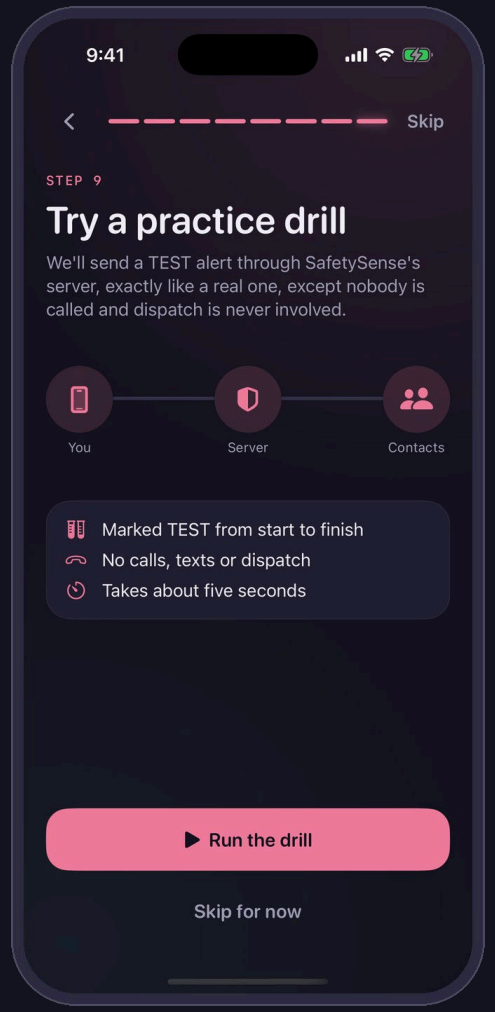
Written to be calm and never blaming. Advocate review is our next step.

Your first alert is a drill

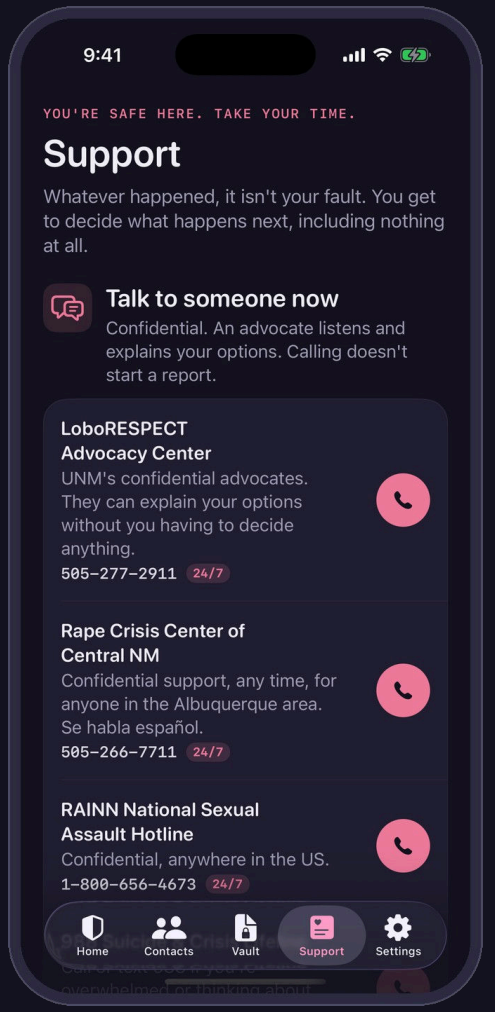
Onboarding ends with a practice run marked TEST.

Afterward, you choose

An advocate, medical care, reporting, or keeping it private.



The practice drill



Support, afterward

PRIVACY AND SECURITY

Private by default. Evidence only when it matters.

1

Guard on

Listening happens on the phone

2

30-second buffer

In memory, overwriting itself

3

Only if an alert fires

Audio is encrypted on the phone before upload

4

Vault

Opens with your Face ID. You decide who hears it

No audio is saved unless an alert fires

No ad or analytics SDKs

Live links are built to expire after the alert

TECHNOLOGY

A real app on a real backend

On the iPhone

- Apple sound, speech and motion models
- Rules-based risk engine
- On-device language model as a second opinion

heartbeat
+ location
no audio while guarding

Our server (Supabase)

- Alerts and the ladder
- Heartbeat watchdog, once a minute

Reaching people

- Twilio: calls and texts to contacts
- Live-link web page
- Noonlight: professional dispatch (sandbox today)

PATHWAY

How we'll know it works, and what we haven't proven yet

TARGETS, NOT YET MEASURED

< 1
false alert per 10 h

< 60 s
scream to ringing phone

< 8%
battery per hour

80%
finish onboarding

Not proven yet

- No field test in real, noisy places
- Real-device run on an iPhone 16e is pending
- Live dispatch and our relay still need Noonlight's sign-off
- Internal TestFlight only

NEXT

Student usability tests and advocate review → **TestFlight pilot** → **Campus partnership**

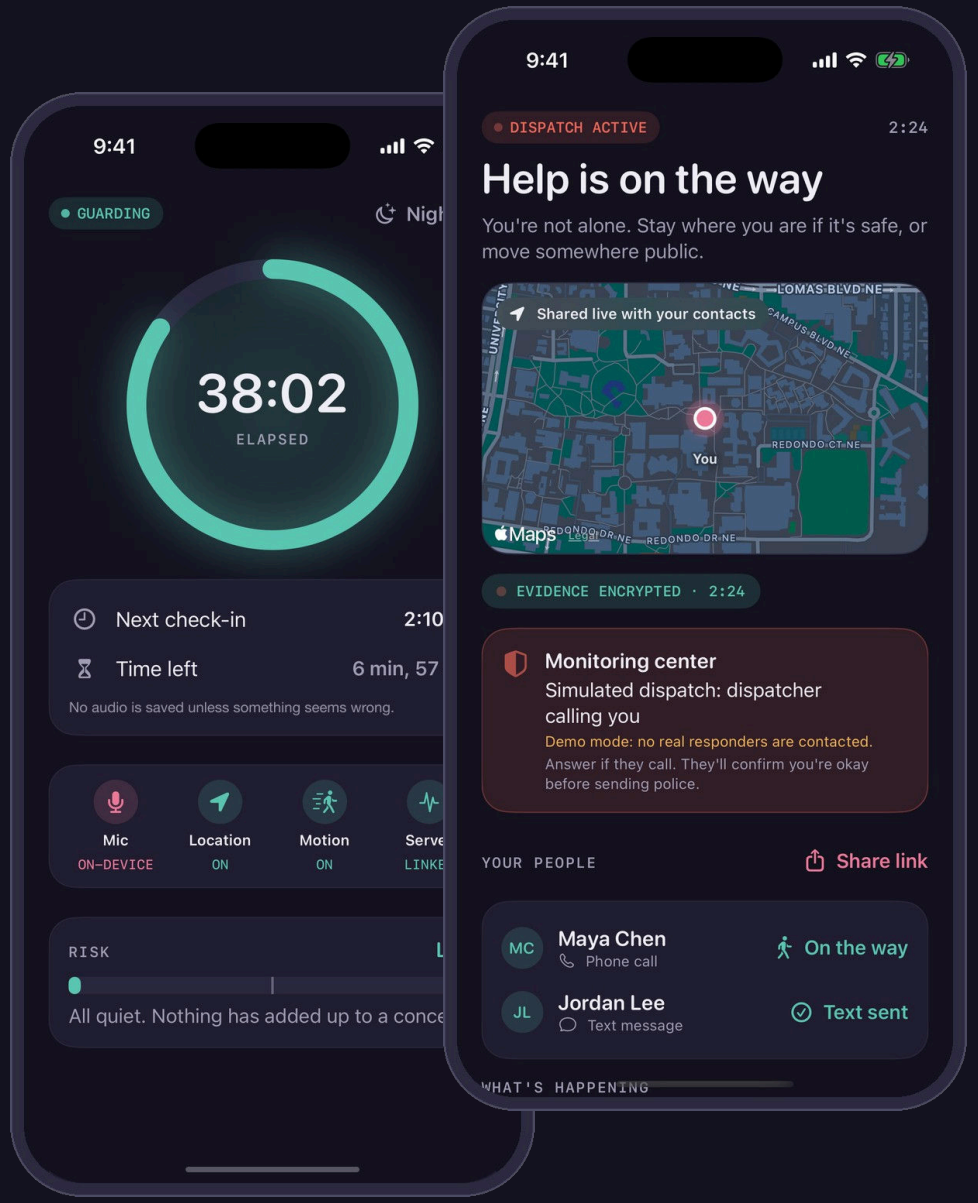
SAFETYSENSE

When you can't call for help, SafetySense does.

SafetySense supplements 911. It never replaces it.

George Teifel UNM **Mary Teifel** CNM

Thank you. We'd love your questions.



Likely judge questions: trust and tech

What about false alarms?

Escalation needs a combination of signals that fade over time, and the first step is always a quiet “You okay?” that one tap clears. Contacts come before dispatch, and the monitoring center's call is forwarded to you to confirm before sending police. Target: under 1 false alert per 10 hours.

Won't listening kill the battery?

It only listens during a Guard session you start, not all day. Sensing runs on Apple's on-device frameworks, and the server heartbeat is one small request a minute. Our target is under 8% per hour. We haven't measured it on a real phone yet; an iPhone 16e run is next.

Is it recording my conversations?

No. Audio sits in a 30-second memory buffer that overwrites itself and is never written to storage. Only an Alert keeps audio, and that's encrypted on the phone. Transcription and AI run on the device. The server sees ciphertext only.

Will Apple approve it?

We follow the rules rather than fight them: listening only during user-started sessions with a visible Live Activity, no auto-dialing 911 (iOS doesn't allow it), and dispatch would go through Noonlight, a professional monitoring service many apps already use. Apple hasn't reviewed it: the app is on internal TestFlight only.

Likely judge questions: cost, law, business

What does it cost to run?

A Twilio call or text costs a cent or two, and only during alerts. Supabase is free at pilot scale and about $25 a month after that. Dispatch is priced per alarm through Noonlight's partner program, which we'd confirm before launch.

Why not Apple Check In or UNM Safe?

Both are good, and we link to UNM Safe. Check In notices when you don't arrive or a timer runs out. UNM Safe needs you to tap. Neither listens for distress or has a duress PIN. We cover the moments you can't reach your phone.

Is recording legal?

Audio is kept only during an alert, and the user is a party to the conversation. New Mexico is generally read as a one-party-consent state (NMSA 30-12-1), but we'd get a legal review before launch. Clips belong to the user and are shared only if they choose.

Business model and partners?

Free for students. Proposed revenue is a campus licence paid from safety and Title IX budgets, plus a paid plan for non-students. Partners we hope to work with: LoboRESPECT, the Women's Resource Center, UNMPD and CNM Security.

Likely judge questions: edge cases

What if nobody answers, or there's no signal?

If no contact responds in 90 seconds, dispatch takes over. If the phone loses signal or dies mid-session, the server notices the missing heartbeats and alerts contacts with the last known location.

Could an abuser use it to track someone?

Location is shared only while you're guarding or in an alert, and only you can start that. Contacts can't pull your location on demand, and live links are built to expire after the alert.

What about older iPhones?

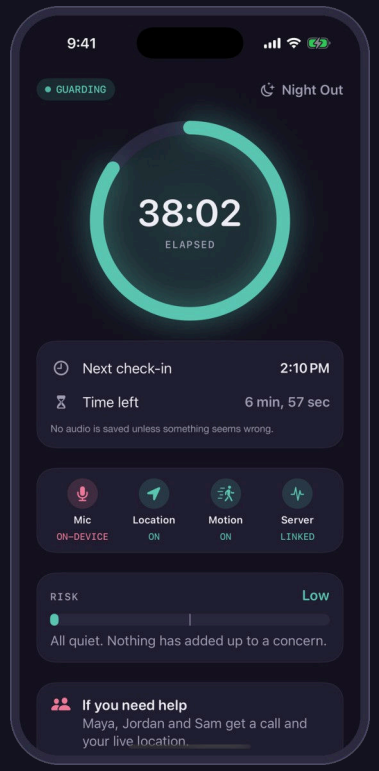
The Action Button needs a newer iPhone, so there's also a Lock Screen control, Siri, a spoken safe word and a hold-to-release SOS. Without Apple Intelligence the app runs on rules only. It needs iOS 26.

Android? Other campuses?

We started with iOS so we could use Apple's on-device AI. The backend is platform-neutral, and the contacts' live link already works in any browser. Android and other campuses come after the student pilot.

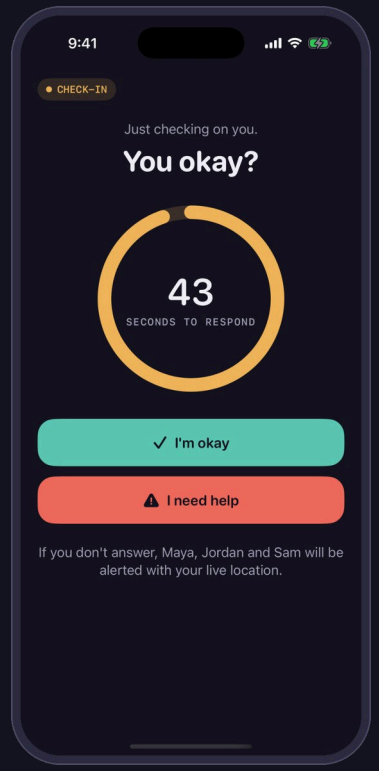
One calm ladder, from “You okay?” to dispatch

0 · Guarding



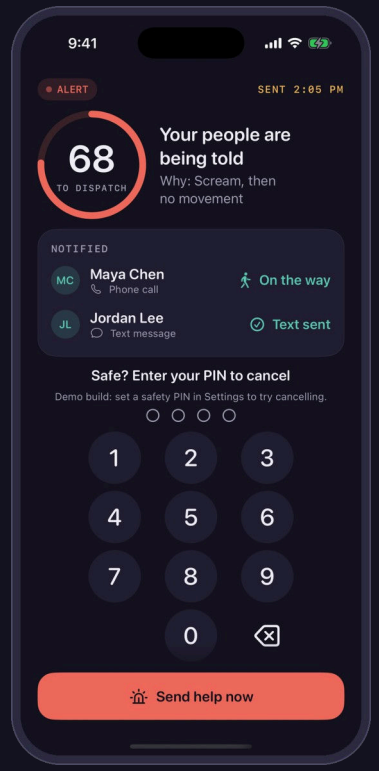
Audio stays in a 30 s memory buffer. No audio is saved.

1 · Check-in



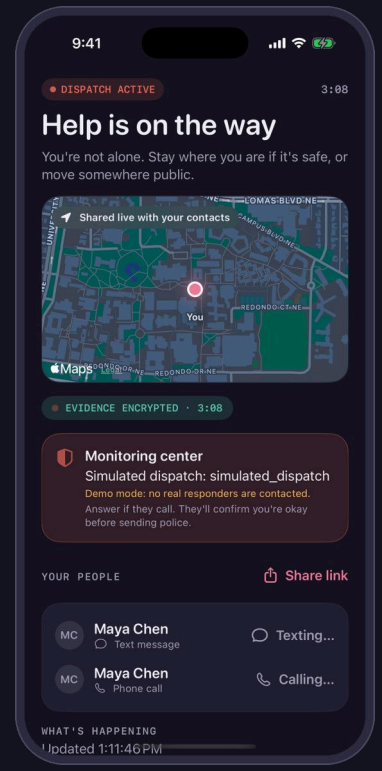
A quiet haptic: “You okay?” You have 45 seconds to tap.

2 · Alert



Your people get a call and a live map. Evidence is encrypted.

3 · Dispatch (simulated)



After 90 s or “Send help,” a monitoring center calls and can send police. Simulated here.